Where Small Quantum Classifiers Help, and Where They Don't: Fair Baselines and Real-Hardware Results on IBM Heron

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Abstract

We compare small quantum neural networks (QNNs) with classical models under a fixed protocol: classical baselines get the same inputs and are tuned by cross-validation, hardware runs are compared with noise-free simulation on identical inputs, and hardware deviations are tested against the exact shot-noise expectation. In a pre-registered evaluation on untouched data, a 4-qubit QNN on sentence-encoder features is within about two points of tuned classical models on the same features and ahead of none (sentiment 0.766 vs. 0.769; CVE severity 0.708 vs. 0.720, a tie within a pre-set ±3-point margin); a 1.5-point gap seen during design did not replicate. An ablation crossing encoding with re-uploading attributes most of the design effect to the encoding (19–24 points). On IBM's `ibm_marrakesh` (200 inputs per task, repeated runs), no run's accuracy differs significantly from noise-free, but the mean output shifts towards one class in every run (-0.002 to -0.006); single runs on two further Heron devices show the same sign with a device-dependent size (to -0.015), which a same-device noise model matched on one device but not the other. In simulation, a trained two-copy QNN separates pure from mixed states with 16–32 copies per state from 2 to 6 qubits, against 128–4,096 for the classical-shadow baseline we test, illustrating a known multi-copy separation; a preliminary 2-qubit hardware run keeps the QNN ahead by a smaller factor (2.2–2.7, against about 4 in simulation). All code, results and job identifiers are archived.

1 Introduction

Claims of quantum advantage in machine learning are hard to evaluate when results are reported without matched classical baselines [1]. We ask a narrow question: for small, trainable quantum classifiers that run on current hardware, on which tasks do they differ from well-tuned classical models, and how much does real-device noise change their outputs?

We report two projects. The first trains a 4-qubit QNN on classical data, including two text-classification tasks, and runs the trained circuits on IBM Quantum hardware. The second classifies quantum states directly, a setting where theory predicts separations for learners with quantum memory [2–4]; its purity result is in exact simulation up to 6 qubits, with a 2-qubit check on hardware. Our contributions are:

- an evaluation protocol that keeps quantum and classical models on identical inputs, tunes the classical baselines, and compares hardware with simulation on identical inputs (Section 2);
- evidence from a pre-registered evaluation on untouched data, with seed variation and paired intervals, that a well-designed QNN is close to but not better than tuned classical models on the same features for two text tasks, and an ablation of the circuit design (Section 3);

- a repeated hardware study on fixed physical qubits, with paired tests against noise-free simulation, showing no significant accuracy change but a small output bias of consistent sign and varying size, replicated in sign on two further devices (Section 4);
- a small-scale simulation of the copy-complexity gap between a learned two-copy QNN and a classical-shadow baseline for purity testing, with per-seed results, measured gradient variance, and a 2-qubit hardware run in which the gap narrows but persists (Section 5).

2 Evaluation protocol

1. **Same inputs.** Classical baselines (logistic regression and an RBF SVM; an MLP where noted) receive exactly the features encoded into the circuit.
2. **Tuned baselines and a classical reference.** Same-feature baselines are tuned by 5-fold cross-validation on the training data only. We also report logistic regression on the full feature set and all training data.
3. **Same inputs on hardware.** Device runs use a fixed random subset of the test set; they are compared only with the noise-free simulator on that subset.
4. **Shot noise as the null hypothesis.** Prediction changes on hardware are compared with the exact distribution expected from finite sampling alone, and repeated in separate jobs.
5. **Pre-registered confirmation.** Because the design was chosen with test-set feedback (below), the final text-task comparison is repeated once on untouched data, with the system, the analysis and the decision rules committed publicly before that data was downloaded.

Disclosure. The circuit design and feature-reduction method were chosen in exploratory runs that reported test-set accuracy; no separate validation set was held out for those choices. The test-set numbers in Table 1 are therefore exploratory; the pre-registered evaluation in Table 2 is the confirmatory result for the text tasks. The ablation and the quantum-data experiments have no such untouched replication. Rule 3 was introduced after our first hardware chart placed 40-input device accuracy beside 200-input simulator accuracy, which made the device appear better than the simulator on one dataset.

3 Quantum classifiers on classical data

Model. Inputs are reduced to one feature per qubit and min–max scaled to $[0, \pi]$ using the training data. The circuit repeats, twice, an $R_Y(x_i)$ encoding layer followed by one trainable `RealAmplitudes` layer (data re-uploading [5, 6]); the class is the measured value of qubit 0 (class 1 if $P(1) > 0.5$). Weights are initialised uniformly in $[-\pi, \pi]$ from the seed and trained with L-BFGS-B (at most 60 iterations, one start) on the cross-entropy of the exact simulated output probabilities, with one-hot labels. We used Qiskit 1.4 [7] and qiskit-machine-learning 0.8.

Text data and partitions. Texts are embedded with all-MiniLM-L6-v2 [8] (384 dimensions) and reduced to four features with partial least squares (PLS), fitted on the full training split; the scalers are also fitted on training data only. *Sentiment*: Rotten Tomatoes sentence polarity [9], the standard 8,530/1,066 train/test split, balanced; the 1,066-sentence validation split was never used until the confirmatory evaluation. *CVE severity*: 12,116 NVD descriptions with CVSS v3.1 scores,

published 1 January–29 April 2024, labelled high/critical if the base score is at least 7.0, and split by publication date: training up to 9 April 2024 (9,692; 49.9% high/critical), test after it (2,424; 42.2%). The label shift explains the 0.578 majority-class baseline. For each seed, the QNN and the same-feature baselines are trained on the same balanced random subset of 600 training texts; the full-embedding reference uses all training texts.

Model (test accuracy, mean ± std over 3 seeds)	Sentiment (1,066)	CVE severity (2,424)
Majority class	0.500	0.578
QNN, 4 qubits	0.757 ± 0.007	0.699 ± 0.015
Logistic regression, same 4 features, defaults	0.770 ± 0.000	0.688 ± 0.004
Logistic regression, same 4 features, CV-tuned	0.772 ± 0.002	0.689 ± 0.004
RBF SVM, same 4 features, CV-tuned	0.770 ± 0.004	0.691 ± 0.008
Logistic regression, full embedding, all data (defaults / CV-tuned)	0.774 / 0.774	0.715 / 0.702
QNN − tuned logistic regression, 95% CI over seeds and items	[−0.029, −0.003]	[−0.013, +0.035]

Table 1: Text tasks, exploratory test sets (population std over seeds). The interval for the difference is a two-level percentile bootstrap that resamples the three seeds and then the test items (10,000 resamples); per-seed paired intervals and exact McNemar tests are in the results files. Tuning grids: $C \in \{0.01, \ldots, 100\}$; for the SVM also $\gamma \in \{\text{scale}, 0.1, 1, 10\}$.

On the exploratory test sets the QNN is about 1.5 points below tuned logistic regression on sentiment (interval excludes zero) and nominally ahead on CVE severity (interval includes zero). Cross-validated tuning of the full-embedding model lowered its CVE test accuracy (0.715 to 0.702), consistent with the label shift between the time-ordered splits; we report the tuned value as the reference.

Confirmatory evaluation. We then froze the system and scored it once on data no design decision had touched: the Rotten Tomatoes validation split, and all 13,432 CVEs with CVSS v3.1 scores published 1 May–28 August 2024 (47.9% high/critical). The protocol, script and decision rules were committed before the data was downloaded (Appendix A). The pre-registered expectations were that the QNN is below tuned logistic regression on sentiment and within ±3 points of it on CVE severity. Table 2 gives the outcome. On sentiment the expected gap did not replicate: the QNN is within 0.3 points of tuned logistic regression, and the interval includes zero. On CVE severity the pre-registered tie is confirmed, but the QNN is now nominally below rather than ahead, and in a secondary contrast it is 1.8 points below the tuned SVM, with an interval that excludes zero. Taken together, on untouched data the QNN is within about two points of tuned classical models on the same four features and is not ahead of any of them.

Which design choice mattered? An earlier design, using the `ZZFeatureMap` encoding [10], parity readout of all qubits and training on 1,024-shot sampled probabilities with COBYLA, scored about 0.69 on breast cancer. Table 3 changes one factor at a time from the final design, over three seeds (the seed sets the 70/30 split and the initial weights), and crosses encoding with re-uploading in a 2×2 design. Replacing angle encoding with `ZZFeatureMap` lowers accuracy by 24 points with re-uploading (0.910 to 0.667) and by 19 points without (0.887 to 0.700), so the encoding effect does not depend on re-uploading; re-uploading itself helps angle encoding (+2.3 points) and slightly hurts `ZZFeatureMap` (−3.3). The other single changes, including 2 or 6 qubits instead of 4, move accuracy by at most 1.4 points. Breast cancer was part of the exploratory benchmark, so these numbers are descriptive.

Confirmatory set (accuracy, mean ± std over 3 seeds)	Sentiment (1,066)	CVE severity (13,432)
Majority class	0.500	0.521
QNN, 4 qubits	0.766 ± 0.003	0.708 ± 0.009
Logistic regression, same 4 features, CV-tuned	0.769 ± 0.006	0.720 ± 0.004
RBF SVM, same 4 features, CV-tuned	0.767 ± 0.002	0.726 ± 0.005
Logistic regression, full embedding, CV-tuned	0.777	0.725
QNN − tuned logistic regression [95% CI]	−0.003 [−0.017, +0.011]	−0.011 [−0.024, +0.003]
QNN − tuned SVM [95% CI]	−0.001 [−0.013, +0.012]	−0.018 [−0.031, −0.003]
Pre-registered verdict	expected gap not confirmed	tie within ±3 points confirmed

Table 2: Pre-registered confirmatory evaluation on untouched data (sample std over seeds; two-level bootstrap intervals over seeds and items).

Variant (breast cancer, 300 training / 171 test, PCA features)	Accuracy	vs. final
Final: angle encoding, re-uploaded twice, qubit-0 readout, exact, L-BFGS-B, 4 qubits	0.910 ± 0.011	–
`ZZFeatureMap` encoding, re-uploaded, rest as final	0.667 ± 0.052	−0.244
No re-uploading (angle encoding)	0.887 ± 0.015	−0.023
`ZZFeatureMap` encoding, no re-uploading	0.700 ± 0.032	−0.211
Parity readout	0.924 ± 0.005	+0.014
COBYLA instead of L-BFGS-B	0.903 ± 0.003	−0.008
Sampled (1,024-shot) training, COBYLA	0.899 ± 0.012	−0.012
Original design (all of the above changes)	0.690 ± 0.051	−0.220
2 qubits / 6 qubits	0.916 / 0.899	+0.006 / −0.012

Table 3: Ablation, mean ± std over 3 seeds (population std); the first four rows form a 2×2 design of encoding and re-uploading; all variants are evaluated with exact probabilities. This is a single dataset and should be read as indicative.

Figure 1 gives the same comparison on three standard datasets and on the two text tasks, using smaller, separate splits (300 training and at most 200 test items, PCA for the tabular data). The QNN is within 0–10 points of the best classical model on each and above logistic regression only on the nonlinear two-moons task, where the RBF SVM is best. The text-task numbers in Table 1 supersede the text-task bars in this figure.

4 Real-hardware behaviour

Setup. We ran the trained sentiment and CVE models (seed 0, Table 1) on IBM's 156-qubit Heron device `ibm_marrakesh` through the free Open Plan, with `SamplerV2`, 1,024 shots and no error mitigation or dynamical decoupling. Each circuit was transpiled once with Qiskit's preset pass manager (optimisation level 1, `seed_transpiler`=0), so every run used the same physical qubits (0–3); the compiled circuit has depth 37 with six CZ gates. At execution time the readout errors of these qubits were 0.006–0.024 and the CZ errors 0.0022–0.0027 (full snapshot and job identifiers in Appendix A). For each task we drew a fixed, stratified random sample of 200 test inputs and ran it as separate jobs: three for CVE severity (two at 16:16 and 16:18 UTC, and one about three hours later at 19:22 UTC) and one for sentiment. The device's reported calibration of these qubits was the same for all three CVE runs, so the later run separates them in time but not across a recalibration.

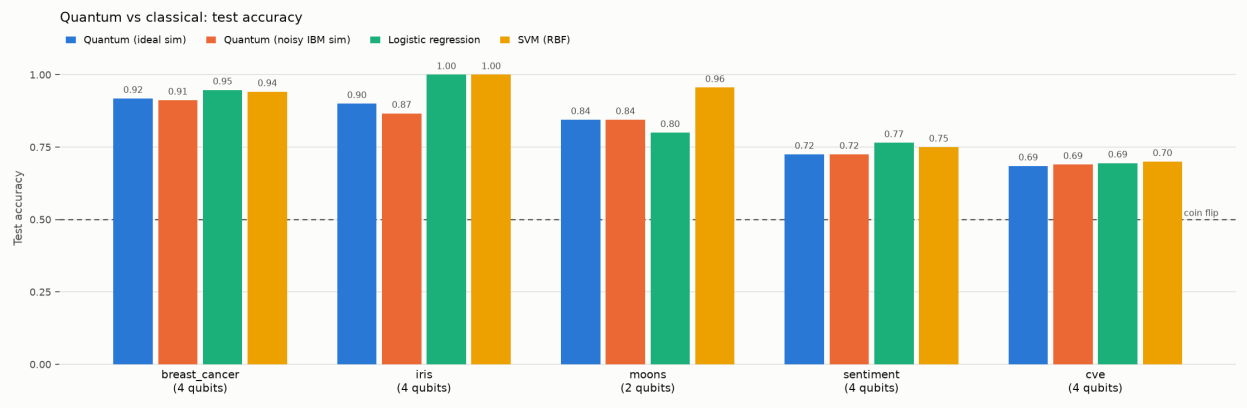


Figure 1: Benchmark on smaller, separate splits: test accuracy of the QNN (noise-free and with the `FakeManilaV2` noise model) and classical models on identical reduced features, one seed.

Statistics. For an input with noise-free probability p and N shots, the probability that sampling alone changes the prediction is the exact binomial tail

$$q(p) = \begin{cases} P(K \le N/2), & p > 1/2, \\ P(K > N/2), & p \le 1/2, \end{cases} \qquad K \sim \mathrm{Bin}(N, p). \tag{1}$$

Under the null hypothesis that the device adds nothing beyond shot noise, every run draws fresh shots, so outcomes are independent across inputs and runs even though runs reuse the same inputs; the total number of changed predictions then follows a Poisson-binomial distribution, which we compute exactly by convolution. Under an alternative (a systematic device effect), outcomes for the same input are correlated across runs, so we also give the test for each run alone. Accuracy on the device is compared with noise-free accuracy on the same inputs by an exact McNemar test per run. Bias is the mean signed difference between device and noise-free $P(\text{class } 1)$, given per run with a 95% bootstrap interval over inputs, and pooled with a two-level bootstrap that resamples runs and then inputs. The same threshold ($P > 0.5$) is used in simulation and on the device.

Results. Figure 2 and Table 4 summarise the study. No run's accuracy differs significantly from noise-free accuracy on the same inputs: the three CVE runs score 0.685, 0.655 and 0.700 against a noise-free 0.685 (exact McNemar $p = 1$, 0.070 and 0.45; discordant pairs 2/2, 1/7 and 5/2), and sentiment scores 0.730 against 0.740 ($p = 0.63$). With 200 inputs this rules out only large accuracy changes. The numbers of changed predictions are at the edge of what shot noise explains: 19 against 13.3 expected over three CVE runs ($p = 0.055$; per run 4, 8 and 7 against 4.4, $p = 0.68$, 0.054 and 0.13), and 4 against 3.1 for sentiment ($p = 0.37$). We read this as compatible with shot noise, not as evidence of its absence. Repeated runs agree on 96% of predictions. The mean output probability shifted towards class 0 in every run. For CVE the per-run shift was -0.0063 $[-0.0090, -0.0037]$ and -0.0053 $[-0.0081, -0.0025]$ in the first session and -0.0022 $[-0.0046, +0.0002]$ three hours later, whose interval includes zero; pooled over the three runs it is -0.0046, with a two-level interval of $[-0.0076, -0.0016]$. For sentiment (one run) it is -0.0035 $[-0.0061, -0.0010]$. The sign was the same in all four runs but the size varied between sessions, even under the same reported calibration, so we report a bias of these executions rather than a stable property of the device; the calibration snapshot records gate and readout error rates and coherence times, not drifts within a calibration cycle. Such a bias can change predictions for inputs near the decision boundary, so label agreement alone understates the effect of noise.

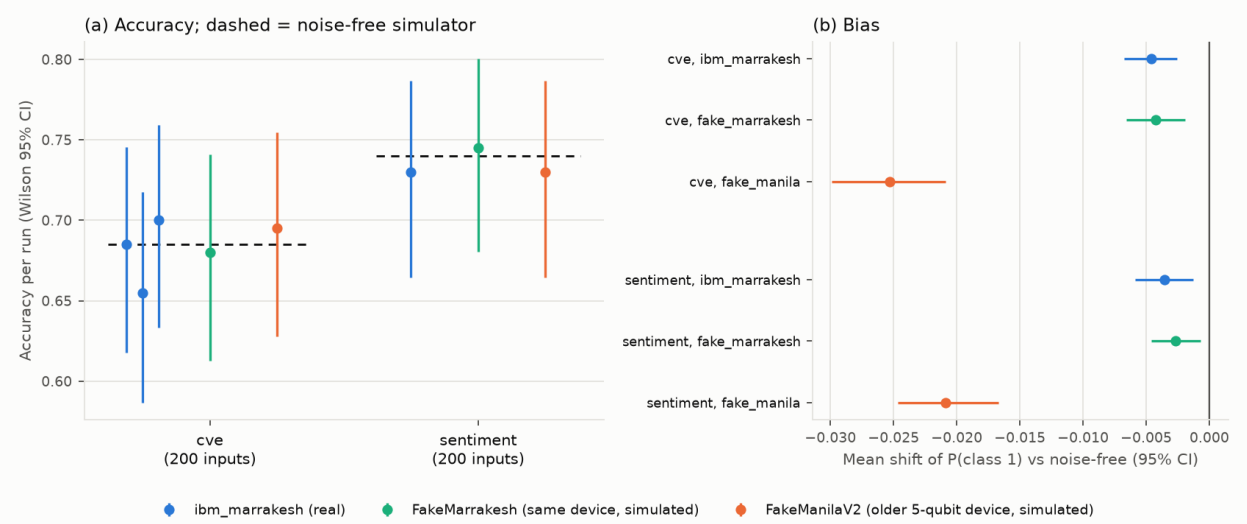


Figure 2: Hardware study, 200 inputs per task. (a) Accuracy per run with Wilson 95% intervals; dashed lines are noise-free accuracy on the same inputs. (b) Mean signed shift of $P(\text{class 1})$ relative to noise-free simulation, pooled over runs, with 95% bootstrap intervals.

Task	Backend	Accuracy per run	Changed / expected	p	Bias [95% CI]
CVE (ideal 0.685)	`ibm_marrakesh`	0.685, 0.655, 0.700	19 / 13.3	0.055	−0.0046 [−0.0076, −0.0016]
	FakeMarrakesh	0.680	3 / 4.4	0.86	−0.0043 [−0.0066, −0.0019]
	FakeManilaV2	0.695	12 / 4.4	< 0.001	−0.0253 [−0.0299, −0.0209]
Sentiment (ideal 0.740)	`ibm_marrakesh`	0.730	4 / 3.1	0.37	−0.0035 [−0.0059, −0.0013]
	FakeMarrakesh	0.745	3 / 3.1	0.63	−0.0027 [−0.0046, −0.0007]
	FakeManilaV2	0.730	8 / 3.1	0.003	−0.0209 [−0.0246, −0.0167]

Table 4: Hardware study, 200 inputs per task, 1,024 shots. "Changed / expected" counts predictions that differ from the noise-free prediction, against the exact shot-noise expectation; $p = P(\text{total} \geq \text{observed})$ under shot noise alone. Bias intervals are 95% bootstraps over inputs (over runs and inputs for the three CVE runs); per-run values are in the text.

Noise models. `FakeMarrakesh`, Qiskit's noise model built from a calibration snapshot of the same device, reproduces both the accuracy behaviour and the bias of the real device within the intervals. `FakeManilaV2`, a model of an older 5-qubit device, overstates the bias about fivefold and changes significantly more predictions than shot noise allows. This compares two specific noise models on one circuit; it suggests matching the noise model to the device, rather than a general statement about noise models, and the next paragraph shows that even a same-device model can miss.

Other devices. The Open Plan gives access to two further 156-qubit Heron r2 devices, `ibm_kingston` and `ibm_fez`. With the remaining quantum time we ran the CVE model once on each, on a 100-input stratified sample that is a subset of the 200 inputs above, compiled in the same way (both used physical qubits 0–3 with the same depth-37, six-CZ circuit). Table 5 compares all three devices on these 100 inputs. No device's accuracy differs significantly from noise-free (exact McNemar $p \geq 0.13$ for every run), and the changed predictions on the two new devices are within shot noise. Every device shifts the output towards class 0, but by different amounts: on the same inputs `ibm_fez` shifts it by 0.010 more than `ibm_marrakesh` (paired 95%

CI $[0.006, 0.015]$) and by 0.007 more than `ibm_kingston` ($[0.002, 0.011]$). `FakeFez` understates the shift on `ibm_fez` by 0.006 ($[0.002, 0.010]$), while `FakeMarrakesh` matches `ibm_marrakesh` (difference $+0.0005$, $[-0.003, +0.004]$); no noise model of `ibm_kingston` is distributed with Qiskit. The sign of the bias therefore replicated across devices, its size did not, and a same-device noise model matched only one of the two devices we could check. One run per new device cannot separate device from session effects.

Backend (100 inputs; noise-free 0.650)	Accuracy	Changed / expected	p	Mean shift [95% CI]
`ibm_marrakesh`, 3 runs	0.650, 0.600, 0.660	4, 7, 5 / 3.1	–	-0.0050 (per run -0.007, -0.007, -0.002)
`ibm_kingston`, 1 run	0.630	2 / 3.1	0.85	-0.0087 [-0.0121, -0.0055]
`ibm_fez`, 1 run	0.640	5 / 3.1	0.17	-0.0152 [-0.0208, -0.0095]
`FakeMarrakesh`	0.660	1 / 3.1	0.97	-0.0045 [-0.0078, -0.0012]
`FakeFez`	0.640	3 / 3.1	0.63	-0.0092 [-0.0135, -0.0049]

Table 5: Cross-device replication of the CVE classifier on 100 identical inputs (a subset of Table 4's inputs); `ibm_marrakesh` rows are its three runs restricted to these inputs. $p = P(\text{changed} \geq \text{observed})$ under shot noise alone.

Earlier, smaller run. A first run on 40 inputs per dataset (30 for iris) on the previous day reproduced the noise-free predictions on four of five datasets; CVE changed three predictions, and three repeats on the same 40 inputs gave 5 changes against 3.45 expected (exact $p = 0.24$; 8 against 4.6, $p = 0.063$, including the first run). These smaller samples motivated the study above.

5 Quantum classifiers on quantum data

The second project gives models direct access to quantum states; classical learners (logistic regression, RBF SVM, MLP) receive classical-shadow estimates [11] from random-Pauli snapshots, or full state vectors where noted. Each experiment uses 200 training and 400 test states per size; entanglement and phase results are over five seeds and purity results over three, each seed redrawing states, shot noise and initialisations. For the purity task, the classical learner and its hyperparameters are selected on a 50-state validation split of the training states and scored once on test. For the entanglement and phase tasks the best classical learner is selected on the test set, which favours the classical side. Throughout, "c copies" means c preparations of the state for every method: the two-copy QNN uses $c/2$ shots of a circuit holding two copies, and the shadow method uses $\lfloor c/9 \rfloor$ shots in each of 9 measurement bases.

Entanglement (2–4 qubits). The task separates product states from entangled states (Meyer–Wallach $Q \geq 0.2$). A two-copy QNN, which receives two copies of each state, is perfect in exact simulation and reaches 0.991 ± 0.006, 0.992 ± 0.004 and 0.996 ± 0.005 at 2, 3 and 4 qubits with 256 copies. A one-copy QNN stays at chance (0.47–0.52 per seed): with one copy and a fixed measurement its output probability is linear in ρ, and this label is not a linear function of the state; this does not imply that every single-copy method fails. Classical learners on full state vectors reach only 0.715 ± 0.018, 0.569 ± 0.024 and 0.524 ± 0.025, but learners on classical-shadow purity features are perfect with exact estimates and reach 0.951 ± 0.009, 0.975 ± 0.005 and 0.988 ± 0.006 with 256 snapshots. The shadow baseline therefore closes the gap to within 0.04 at 2 qubits and 0.01 at 4 qubits: this task does not separate the two approaches.

Phases. For ground states of the transverse-field Ising chain (4–8 qubits) and a $\mathbb{Z}_2$ lattice gauge model (8 link qubits), we use a QCNN with hardware-efficient units: convolution layers of $R_Y R_Z$, CX, $R_Y R_Z$ on neighbouring qubits (8 weights per layer, shared across the ring) and coherent pooling that halves the active qubits down to one readout qubit (4 weights per layer), following the layer structure of [12]. (The 3-parameter units of the Qiskit tutorial output exactly 0.5 on these spin-flip-symmetric states.) The QCNN, a generic QNN and classical models all score 0.96–1.00 overall. These tasks are easy: logistic regression on a *single* exact feature (mean nearest-neighbour ZZ, or mean X) reaches 0.994 ± 0.005 at 4 qubits and 0.995 ± 0.006 at 8, and the mean electric field alone reaches 0.998 ± 0.002 on the gauge model. All errors fall near the critical point (about 20% of test states). There, the exact-simulation QCNN reaches 0.963 ± 0.037 (4 qubits, $|h-1| < 0.2$) and 0.974 ± 0.033 (8 qubits), against 0.901 ± 0.033 and 0.965 ± 0.014 for classical learners on exact Z/X expectations; with spreads of this size, only the 4-qubit difference is suggestive. The phase tasks do not discriminate between models.

Purity. Haar-random pure states are distinguished from random rank-2 mixed states, a global property. Single-copy strategies need a number of copies that grows exponentially with system size, while a two-copy SWAP-type test needs a constant number [2, 3]. The two-copy QNN reaches 1.000 in exact simulation at every size; the one-copy QNN stays at chance (0.44–0.57 per seed). Table 6 reports the copies of each test state needed for 90% test accuracy, on the grid $2, 4, \ldots, 4096$: 16–32 for the QNN's seed-averaged curve (16–64 for individual seeds) against 128–4,096 for the shadow baseline, and the per-seed ranges never overlap. The shadow baseline's learner (logistic regression, RBF SVM or MLP, each over a hyperparameter grid) is chosen on validation data; choosing the best learner on the test set instead gives the same thresholds except at 5 qubits (1,024 rather than 2,048).

Qubits	2	3	4	5	6
Two-copy QNN, mean curve	32	16	32	32	32
Two-copy QNN, per seed	64, 16, 16	16, 16, 16	32, 16, 16	16, 32, 16	64, 32, 16
Classical shadows (validated learner), mean curve	128	256	512	2,048	4,096
Classical shadows, per seed	128 (all)	256 (all)	512, 1024, 512	2048, 2048, 1024	4096 (all)

Table 6: Copies of each test state needed to reach 90% test accuracy, exact simulation. "Mean curve": smallest grid value where accuracy averaged over three seeds is at least 0.9; "per seed": the same for each seed. The one-copy QNN never reaches 90%.

Trainability. The two-copy model was trained at 2 qubits and its shared block transferred to larger sizes with 30 fine-tuning steps. Training with a separate block per copy pair from a random start failed at 5 qubits in one of three seeds (0.85 ± 0.26). To test whether this reflects a barren plateau [13, 14], we measured the variance of the gradient of the class-separation signal $D = \overline{P}_{\text{mixed}} - \overline{P}_{\text{pure}}$ at random initialisations (100 draws $\times$ 3 seeds, exact gradients, 2–7 qubits). With per-pair blocks the variance decays from 9.5×10^{-4} to 1.0×10^{-6}, a factor of 3.9 per qubit (fitted $\log_2$ slope -1.97, 95% CI over seeds $[-2.06, -1.87]$; per seed -1.95, -1.94, -2.01; close to 4^{-n}): exponential-like decay over the measured range, consistent with the barren plateaus expected for this global cost, though six sizes cannot establish the asymptotic rate. With one block shared across pairs the decay slows to a factor of 1.7 per qubit (2.6×10^{-3} to 1.6×10^{-4}; slope -0.80, 95% CI over seeds $[-1.26, -0.47]$). No initialisation in either model had a gradient norm below 10^{-3} in this range. The decay is a plausible cause of the failed seed, and weight sharing coincided with successful training, but we did not test the causal link directly.

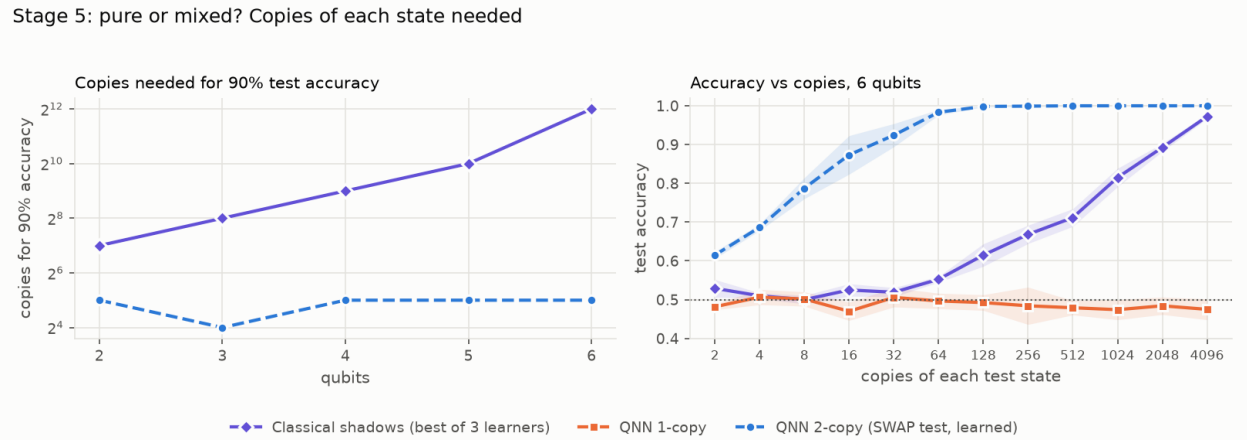


Figure 3: Pure vs. mixed states, exact simulation. Left: copies needed for 90% accuracy (mean curves). Right: accuracy vs. copies at 6 qubits (mean and spread over 3 seeds).

Purity on hardware (preliminary). We ran the 2-qubit purity test on `ibm_marrakesh` as two separate jobs on the same 60 test and 20 calibration states, with 1,024 copies per state and fixed physical qubits. Training (200), test (60) and calibration (20) states are drawn independently and balanced; no calibration state is within trace distance 0.3 of any training or test state. Mixed states are prepared as 3-qubit purifications with the ancilla unmeasured, so that pure and mixed preparations both compile to 7 CZ gates and circuit depth does not reveal the label. After routing, the two-copy circuit has about 19 CZ gates against 7 for a shadow circuit. Models are trained in simulation; on the device each method's decision rule is then refitted on the 20 calibration states ("recalibrated"): a two-parameter logistic regression on the QNN output, and the same on the shadow method's global-purity estimate, so both get the same flexibility. Recalibration changes accuracy by at most about 0.03 at any copy count, and resampling which states are used for calibration widens the intervals but does not change the ordering. Smaller copy counts are subsampled without replacement from the device shots (30 subsamples per point).

At 1,024 copies every model is at 0.99–1.00. Figure 4 shows accuracy against copies. Interpolating each recalibrated curve, the QNN reaches 90% at 45 $[33, 56]$ and 53 $[42, 69]$ copies in the two jobs and the shadow method at 121 $[102, 157]$ and 117 $[103, 138]$, a ratio of 2.7 $[2.0, 4.0]$ and 2.2 $[1.6, 2.9]$ (95% bootstrap intervals over test states and subsamples). In noise-free simulation at this size the grid thresholds are 32 and 128, a ratio of about 4. On this sample the QNN therefore keeps an advantage on the device, by a smaller factor than in simulation. With two jobs on one set of 60 states this is an illustration, not an estimate of hardware scaling. Both methods' purity estimates correlate with the true $\mathrm{Tr}\,\rho^2$ ($r = 0.986$ for the QNN, 0.975–0.981 for shadows) and both read low by 0.06–0.10.

We compared the device with `FakeMarrakesh` by replaying the identical compiled circuits (same physical qubits, gates and shots, downloaded from the device jobs). For these circuits the noise model understates the QNN's output shift: +0.038 and +0.041 on the device against +0.022 and +0.019 in the noise model, and the noise model gives a larger ratio (2.7–3.2). For the shadow circuits device and model agree within the run-to-run spread. The two-copy circuit's greater depth is a plausible reason, but we did not vary depth in a controlled way, so this finding is limited to these compiled circuits and runs. For the shallow text classifier of Section 4 the same noise model matched the device.

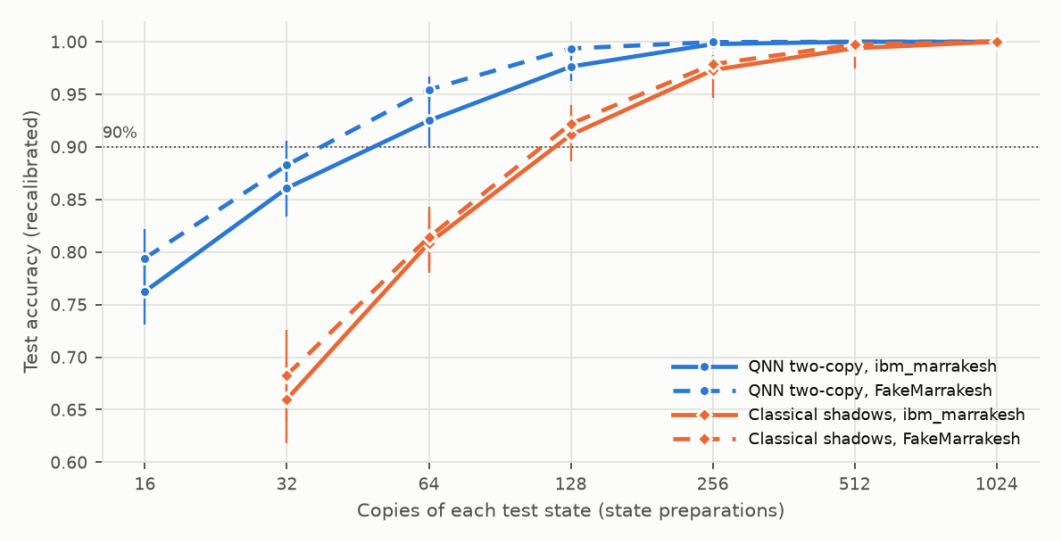


Figure 4: Pure vs. mixed, 2 qubits, on `ibm_marrakesh` (solid) and `FakeMarrakesh` replaying the identical circuits (dashed): recalibrated test accuracy against copies of each test state, subsampled from 1,024-copy jobs. Points are the mean of two jobs; bars span both jobs' 95% bootstrap intervals over the 60 test states. Shadows: logistic regression on the global-purity estimate. At 16 copies the shadow method has one shot per basis and was not evaluated.

Entanglement and phases on hardware. The 2-qubit entanglement and 4-qubit Ising classifiers ran as two jobs each on 40 test and 16 calibration states with 512 copies (Table 7). The two-copy QNN loses 20 points on the raw device output and recovers to 0.988 ± 0.018 after recalibration; its output shift ($+0.028$, $+0.032$) is again about twice that of `FakeMarrakesh` replaying the identical circuits ($+0.016$, $+0.015$). For the shallower QCNN device and noise model agree (-0.007 and -0.009 in both). With 80 pooled predictions a Wilson 95% interval around 0.95 spans about $[0.88, 0.98]$, so no ranking within a task is significant.

Task	Model	Noise-free	Device	Recalibrated
Entanglement, 2 qubits	Two-copy QNN	1.000	0.800 ± 0.035	0.988 ± 0.018
	Best classical (snapshots)	1.000	0.938 ± 0.018	0.975 ± 0.000
Ising phases, 4 qubits	QCNN	0.950	0.925 ± 0.035	0.925 ± 0.035
	Best classical (snapshots)	0.975	0.962 ± 0.018	0.962 ± 0.018

Table 7: Quantum-data classifiers on `ibm_marrakesh`: test accuracy on 40 states, mean $\pm$ std over two runs, 512 copies. "Best classical" is the best of three learners in each column, which favours the classical side.

6 Related work

Benchmarking. Bowles et al. [1] benchmark QML models at scale and find that classical baselines often match or beat them; our protocol follows the same spirit at small scale, adding hardware. **Encoding and expressivity.** Data re-uploading [5] and the Fourier view of encoding [6] explain why the encoding dominated our ablation. **Dequantization.** Quantum-inspired classical algorithms [15] and random-Fourier-feature approximations of variational QML models [16] suggest

that circuits like ours may be classically approximable; we did not test such surrogates. **Learning from quantum data.** Exponential separations between learners with and without quantum memory are proven in [3, 4] and demonstrated on hardware in [2]; our purity experiment is a small-scale illustration of that separation against one classical-shadow method [11], in simulation and at 2 qubits on hardware. **Trainability.** Barren plateaus [13] and their dependence on the cost function [14] predict the exponential gradient decay we measured for per-pair blocks. **Hardware characterisation.** Capability benchmarks [17] and error mitigation [18] address device noise systematically; we used no mitigation beyond refitting decision rules on labelled device states, and characterise only output bias and accuracy for a few small circuits.

7 Discussion and limitations

In these experiments the quantum model used fewer copies than the classical baseline only in the purity task, where the tested two-copy QNN could hold two copies of a state at once, which matches theory [2, 3]; we illustrate that result rather than extend it, and only against the classical-shadow baseline we tested. On hardware that advantage shrank, because the circuit that exploits two copies is deeper, and a same-device noise model did not anticipate how much. On classical data the circuit classifies features chosen by a classical encoder, and tuned classical models on the same features do as well or slightly better.

Limitations: (i) all simulations use at most 8 qubits and are exact, so nothing here is classically hard; (ii) design choices were explored with test-set feedback (Section 2), which the confirmatory evaluation addresses only for the two text tasks; (iii) the ablation uses one dataset, itself part of the exploratory benchmark; (iv) the hardware study uses one circuit family on three devices of one processor family, with 200 inputs and few runs on one device and a single 100-input CVE run on each of the others, and the bias may drift with calibration; (v) prediction changes are tested under an independence assumption across inputs and runs; (vi) CVE severity from descriptions alone approximates CVSS scoring; (vii) the purity test ran on hardware only at 2 qubits and in two runs (preparation circuits for 3 or more qubits were too deep for useful fidelity, and no budget remained for a third run); (viii) the quantum-data hardware runs use 40–60 test states, too few to rank models within a task.

Reproducibility. Code, configurations, result files and job identifiers are archived on Zenodo as AGI v1.0.2 [19] (git commit `2c46075`) and Entangle v1.0.0 [20] (git commit `c4d3528`), with the DOIs given in the references; every number in this paper comes from these versions. Appendix A lists software versions, commands and jobs. During this work we found that `qiskit-ibm-runtime` 0.44.1 declares Qiskit 1.x support but fails to transpile for real devices on it; we reported it upstream (Qiskit/qiskit-ibm-runtime#3563).

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A Reproducibility details

Software. Python 3.12.16; qiskit 1.4.6; qiskit-ibm-runtime 0.43.1 (0.44.x fails on Qiskit 1.x); qiskit-machine-learning 0.8.4; qiskit-aer 0.18.2; numpy 2.4.6; scikit-learn 1.9.1; sentence-transformers 6.2.0. Training uses Qiskit's exact reference `Sampler`.

Code versions. AGI v1.0.2, commit `2c460757ecc6`, https://doi.org/10.5281/zenodo.24149645; Entangle v1.0.0, commit `c4d3528ea23d`, https://doi.org/10.5281/zenodo.24148954. Result-file paths below are relative to these versions.

Commands (AGI repository): `python train.py -task sentiment|cve`; `python -m benchmark.uncertainty` (Table 1); `python -m benchmark.ablation` (Table 3); `python -m benchmark.confirmatory` (Table 2; protocol `results/confirmatory/PROTOCOL.md`, committed as `dda5f41` on 9 November 2026 at 19:37 UTC, before the confirmatory data was downloaded); `python -m benchmark.hardware_study` and `benchmark.hardware_study_report` (Table 4; `-fake-only` reproduces the noise-model rows without IBM access); `python -m ibmq.job_log`. Result files are in `results/`.

IBM jobs (Open Plan; times UTC). Cross-device runs, 12 November 2026: `ibm_kingston` `dbqhgr8k09pc73at9aig`, `ibm_fez` `dbqhh8gk09pc73at9bc0` (29 quantum seconds each; `results/hardware_study/cross_device.md`). All others on `ibm_marrakesh`. 200-input study, 9 November 2026: CVE `dbov06tt7fqc7388gab0`, `dbov3ttt7fqc7388gfrg`, `dbp1pvgou4as73cst2n0`; sentiment `dbov4itt7fqc7388ggog`. Earlier 40-input runs, 8–9 November 2026: benchmark `dbolcblt7fqc73881hv0`, `dbold1tt7fqc73881iqg`, `dboldboou4as73cs9lcg`, `dbole58qtr0c738d1ccg`, `dbolhc0qtr0c738d1ihg`; CVE repeats `dbomf5gou4as73csbdmg`, `dbomfas4148s739t8q1g`, `dbomfg5t7fqc73883ctg`. The full list with timestamps and quantum seconds is in `results/ibm_jobs.json`.

Quantum-data jobs (Entangle repository, `ibm_marrakesh`, 9 November 2026): purity `dbp0rd0ou4as73csr3t0`, `dbp0s1lt7fqc7388jap0` (pilot `dbp0r10qtr0c738dj0fg`); entanglement and Ising, run 1 `dbp11agqtr0c738djefg`, `dbp11eoou4as73csrgp0`, `dbp11mc4148s739tpbpg`, `dbp11rlt7fqc7388jns0`; run 2 `dbp124gqtr0c738djg4g`, `dbp128k4148s739tpcv0`, `dbp12g5t7fqc7388jp20`, `dbp12lc4148s739tpdm0`. Summary with Wilson intervals: `results/review_hardware.md`; seed tables `results/review_seeds_stage*.md`; validated learner selection `results/review_selection.md`; gradient variance and slope intervals `results/review_gradients.md`; hardware purity recalibration, copy accounting and intervals `results/review_purity_hw.md`; per-qubit calibration of the device and of `FakeMarrakesh` `results/review_calibration.md`. Raw device counts and the compiled circuits of every job are in `results/ibm_raw/`; recomputing from them reproduces every device number, and the `FakeMarrakesh` comparisons replay those circuits. Qubits used: median T_1 205–236 μs, T_2 136–152 μs, readout error 0.007–0.018, CZ error 0.0021.

Calibration (physical qubits 0–3 at the start of the 200-input study): readout error 0.006–0.024; CZ error 0.0022–0.0027; full values per qubit, including T_1 and T_2, in `results/hardware_study/*.json`.